

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular & Supplementary Winter Examination – 2024**

Course: B.Tech. Branch : Mechanical Engineering / Mechanical Engineering Sandwich / Automation & Robotics/Robotics/ Mechatronics Engineering

Subject Code & Name: BTMOE505B

Renewable Energy Sources

Semester : V

Max Marks: 60

Date: 22/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is a benefit of solar energy? a. Reduction in greenhouse gas emissions b. Low maintenance costs c. Energy independence d. All of the above	1	1
2	Which component of a solar panel converts sunlight into electricity? a.) Inverter b. Battery c. Semiconductor material d. Reflector	1	1
3	In a solar collector, why is the transparent cover provide for? a. Protect the collector from dust. b Reduce the heat losses from collector beneath to atmosphere c. Transmit solar radiation only d . All of the above.	1	1
4	An instrument used to measure total solar radiation is called a. Hygrometer b. Pyranometer c. anemoter d. Pyrliometer	2	1
5	Which component of a solar panel converts sunlight into electricity? a.) Inverter b. Battery c. Semiconductor material d. Reflector	2	1
6	Maximum efficiency is obtained in.	2	1

	a. Flat plate collector	b Evacuated tube collector	c Line focussing collector	d Paraboloid dish collector		
7	The wind speed is measured using an instrument called				3	1
	a. hydrometer	b Manometer	c anemometer	d. wind vane		
8	A controller in commercial type wind turbines are provided with data from				3	1
	a. Anemoter	b. Yaw drive	C .Rotor	d. None		
9	Which energy forms can biomass be converted to?				4	1
	a.) Electrical and light	b. Light and chemical	c. Electrical and heat	d. Heat and light		
10	What is biomass?				4	1
	a. Organic materials from living organisms	b.) Inorganic materials from living organisms	c. Inorganic materials from non-living organisms	d. Organic materials from non-living organisms		
11	What does OTEC stand for?				5	1
	a) Ocean thermal energy cultivation	b) Ocean thermal energy conversion	c) Ocean techno energy conservation	d) Ocean thermal energy consumption		
12	How is OTEC caused?				5	1
	a. By wind energy	b) By geothermal energy	c. By solar energy	d. By gravitational force		
Q. 2	Solve the following.					12
A)	Explain the difference between renewable and non-renewable energy with suitable examples. Why is renewable energy important?				1	6
B)	Explain in details Energy reserves in India				1	6

Q.3	Solve the following.		12
A)	Explain with neat sketch construction and working of pyroheliometer.	2	6
B)	Define a) Declination angle b) Solar Azimuth angle c) Altitude angle.	2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain with neat sketch construction of flat plate collectors & list out its limitation	2	6
B)	Explain construction and working of Photo Voltaic cell	3	6
C)	Explain Solar drying system with advantages & disadvantage	3	6
Q.5	Solve Any Two of the following.		12
A)	What is wind energy? Give classification of types of wind mills	4	6
B)	Explain selection of site for wind turbine generation system.	4	6
C)	What is biomass energy? Explain with neat sketch floating drum type	4	6
Q. 6	Solve Any Two of the following.		12
A)	Write a short note on Geothermal energy & OTEC system	5	6
B)	Explain Nuclear Reactor	5	6
C)	What is OTEC system? Explain working of open cycle OTEC system.	5	6
	*** End ***		